

RESUME SCREENER & RANKING SYSTEM

The ResumelQ Pro system uses a sophisticated machine learning and NLP pipeline to evaluate candidates. Here is a clear explanation of how the system scores resumes, why some candidates rank higher than others, and how missing skills are identified.

How Resumes are Scored

The system assigns each candidate a **Composite Score (0–100)**, which serves as the final ranking metric 1. This score is not a simple keyword count but a weighted combination of three distinct factors:

- **TF-IDF Textual Similarity (55% weight):** This is the most significant component of the score 1. The system uses **TF-IDF (Term Frequency-Inverse Document Frequency)** to evaluate the importance of words in both the resume and the job description (JD) 2. It then calculates the **Cosine Similarity** between the two to determine the semantic textual overlap, scored from 0 to 1 2.
- **Technical Skill Overlap (35% weight):** The system extracts technical skills from both the JD and the resume using **spaCy** and predefined keyword lists 1, 2.
- **Soft Skill Overlap (10% weight):** Similar to technical skills, the system identifies and weights soft skills to provide a more holistic view of the candidate 1.

Why Certain Candidates Rank Higher

Candidates rank higher when their resumes demonstrate a high degree of alignment with the specific requirements of the job description. Key factors include:

- **Strong Semantic Alignment:** Because textual similarity carries a 55% weight, candidates who use language that closely mirrors the JD's context and terminology will score significantly higher 1, 2.
- **Technical Proficiency:** Since technical skills account for over a third of the score, candidates possessing the exact tools, languages, or platforms mentioned in the JD gain a major advantage 1.
- **"Strong Fit" Threshold:** Candidates who achieve a **composite score of 70 or higher** are automatically labeled as a **"Strong Fit"** for the role 3.
- **Effective Preprocessing:** The system uses **NLTK** to lowercase, remove punctuation, and **lemmatize** words (e.g., converting "running" to "run") 4. Candidates who use standard professional terminology that the system can easily lemmatize and match will be more accurately ranked 4.

Identifying Missing Skills

The system performs a **Skill Gap Identification** to help recruiters understand what a candidate lacks despite their overall score 5. This is achieved through two primary methods:

- **Direct Comparison:** The engine cross-references the skills extracted from the job description against those found in the candidate's resume 1. Any crucial skills required by the JD that are absent from the resume are flagged as **"Missing Skills"** 1.

- **Dataset-Driven Verification:** The system can also verify a resume against a broader **Kaggle resume dataset** 3. It predicts the job class (e.g., "Data Scientist") and compares the candidate's keywords against **industry-standard keywords** for that specific class 3, 6. This allows the system to identify "missing keywords" that are generally expected for that type of role, even if they weren't explicitly stressed in the specific JD 6.